

WBJEE MOCK TEST: PHYSICS

Total Marks: 100 | Time Allowed: 2 Hours

General Instructions

1. This paper contains 75 questions divided into three categories
2. **Category-I (Q1 - Q50):** 1 Mark each. Only one option is correct. Negative marking is -0.25 marks.
3. **Category-II (Q51 - Q65):** 2 Marks each. Only one option is correct. Negative marking is -0.5 marks.
4. **Category-III (Q66 - Q75):** 2 Marks each. One or more options may be correct. No negative marking. Partial marks are awarded for partially correct answers.

CATEGORY - I (1 Mark Each)

Only one option is correct. Correct response yields 1 mark. Incorrect response yields -0.25 marks.

1. The dimensional formula for Planck's constant is identical to that of:

- (A) Linear momentum
- (B) Angular momentum
- (C) Energy
- (D) Power

2. A block of mass m is placed on a smooth inclined plane of inclination θ . The inclined plane is given an acceleration a horizontally so that the block does not slide down. The value of a is:

- (A) $g \tan\theta$
- (B) $g \sin\theta$
- (C) $g \cos\theta$
- (D) $g \cot\theta$

3. In a thermodynamic process, the pressure of a fixed mass of gas is changed such that the gas releases 20 J of heat and 8 J of work is done on the gas. If the initial internal energy was 30 J, the final internal energy is:

- (A) 18 J
- (B) 42 J
- (C) 58 J
- (D) 2 J

4. Two point charges $+q$ and $-q$ are placed at a distance $2l$ apart. The electric potential at a point on the equatorial line at a distance r from the center of the dipole is:

- (A) Zero
- (B) $p / (4\pi\epsilon_0 r^2)$
- (C) $p / (4\pi\epsilon_0 r^3)$
- (D) $2p / (4\pi\epsilon_0 r^2)$

5. A ray of light passes through an equilateral glass prism such that the angle of incidence equals the angle of emergence. If the angle of emergence is $3/4$ times the angle of the prism, the refractive index of the glass is:

- (A) $\sqrt{2}$
- (B) $\sqrt{3}$
- (C) 1.5
- (D) 1.33

6. The half-life of a radioactive substance is 20 minutes. The time taken between 20% and 80% decay will be:

- (A) 20 minutes
- (B) 30 minutes
- (C) 40 minutes
- (D) 60 minutes

7. The ratio of the radii of gyration of a circular disc about a tangential axis in its plane and of a circular ring of the same radius about a tangential axis in its plane is:

- (A) $\sqrt{5} : \sqrt{6}$
- (B) $2 : 3$
- (C) $\sqrt{2} : \sqrt{3}$
- (D) $1 : \sqrt{2}$

8. An electromagnetic wave is propagating along the $+z$ axis. The direction of the electric field (E) and magnetic field (B) can be represented by:

- (A) E along $+x$, B along $+y$
- (B) E along $+y$, B along $+x$
- (C) E along $+z$, B along $+x$
- (D) E along $+x$, B along $+z$

9. A particle is projected with a velocity u making an angle θ with the horizontal. The magnitude of its velocity at the highest point of its trajectory is:

- (A) $u \sin\theta$
- (B) $u \cos\theta$
- (C) u
- (D) Zero

10. A car of mass m is moving on an unbanked circular track of radius R . If the coefficient of static friction is μ , the maximum safe speed is:

- (A) $\sqrt{\mu R g}$
- (B) $\mu R g$
- (C) $\sqrt{\mu g / R}$
- (D) $\mu g / R$

11. A spring of force constant k is cut into two equal halves. The force constant of each half is:

- (A) $k/2$
- (B) k
- (C) $2k$
- (D) $4k$

12. The escape velocity from the Earth is v_e . If a planet has twice the radius and twice the mean density of Earth, its escape velocity will be:

- (A) v_e
- (B) $2 v_e$
- (C) $2\sqrt{2} v_e$
- (D) $4 v_e$

13. The ratio of specific heats $\gamma = C_p/C_v$ for a diatomic gas at normal temperature is:

- (A) 1.67
- (B) 1.40
- (C) 1.33
- (D) 1.28

14. Water rises in a vertical capillary tube up to a height h . If the tube is inclined at an angle of 45° with the vertical, the length of the water column will be:

- (A) h
- (B) $h / \sqrt{2}$
- (C) $h \sqrt{2}$
- (D) $2h$

15. The magnetic flux linked with a coil varies with time t as $\phi = 3t^2 + 4t + 9$ Wb. The magnitude of the induced emf at $t = 2$ s is:

- (A) 8 V
- (B) 16 V
- (C) 22 V
- (D) 32 V

16. An ideal gas undergoes an isothermal expansion at temperature T. The work done by n moles of the gas in expanding from V_1 to V_2 is:

- (A) $nRT \ln(V_2/V_1)$
- (B) $nRT \ln(V_1/V_2)$
- (C) $nRT(V_2 - V_1)$
- (D) $(nR/T) \ln(V_2/V_1)$

17. The efficiency of a Carnot engine working between temperatures of 127°C and 27°C is:

- (A) 25%
- (B) 50%
- (C) 75%
- (D) 100%

18. The distance between two consecutive nodes in a stationary wave is:

- (A) λ
- (B) $\lambda/2$
- (C) $\lambda/4$
- (D) 2λ

19. According to Stokes' law, the terminal velocity of a small spherical body of radius r falling through a viscous medium is proportional to:

- (A) r
- (B) r^2
- (C) r^3
- (D) $1/r$

20. The dimensions of Young's modulus of elasticity are the same as those of:

- (A) Force
- (B) Pressure
- (C) Strain
- (D) Work

21. The moment of inertia of a solid sphere of mass M and radius R about its diameter is:

- (A) $(2/5) MR^2$
- (B) $(2/3) MR^2$
- (C) $(1/2) MR^2$
- (D) MR^2

22. According to Kepler's third law, the square of the time period of revolution of a planet is proportional to the:

- (A) semi-major axis
- (B) square of the semi-major axis
- (C) cube of the semi-major axis
- (D) inverse square of the semi-major axis

23. The electric field outside an infinite thin uniformly charged planar sheet with surface charge density σ is:

- (A) σ / ϵ_0
- (B) $\sigma / (2\epsilon_0)$
- (C) $2\sigma / \epsilon_0$
- (D) Zero

24. The equivalent capacitance of two capacitors C_1 and C_2 connected in series is:

- (A) $C_1 + C_2$
- (B) $(C_1 C_2) / (C_1 + C_2)$
- (C) $(C_1 + C_2) / (C_1 C_2)$
- (D) $|C_1 - C_2|$

25. The drift velocity of free electrons is v_d when current I flows. If both the radius and the current are doubled, the drift velocity will be:

- (A) $v_d / 4$
- (B) $v_d / 2$
- (C) v_d
- (D) $2 v_d$

26. Kirchhoff's voltage law (KVL) is a consequence of the law of conservation of:

- (A) Charge
- (B) Momentum
- (C) Energy
- (D) Angular momentum

27. The magnetic field B at a distance r from a long straight wire carrying current I is proportional to:

- (A) $I \cdot r$
- (B) I / r^2
- (C) I^2 / r
- (D) I / r

28. The force experienced by a charge q moving with velocity v in a uniform magnetic field B is:

- (A) $q(v \cdot B)$
- (B) $q(v \times B)$
- (C) $q(B \times v)$
- (D) $(v \times B) / q$

29. The resonant frequency f_r of a series LCR circuit is:

- (A) $1 / (2\pi\sqrt{LC})$
- (B) $1 / \sqrt{LC}$
- (C) $2\pi\sqrt{LC}$
- (D) $(1/2\pi) \sqrt{L/C}$

30. The core of a transformer is laminated to reduce energy loss due to:

- (A) Eddy currents
- (B) Hysteresis
- (C) Flux leakage
- (D) Copper loss

31. Which of the following electromagnetic waves has the longest wavelength?

- (A) X-rays
- (B) Ultraviolet rays
- (C) Microwaves
- (D) Radio waves

32. A convex mirror always produces an image that is:

- (A) Real, inverted, and magnified
- (B) Virtual, erect, and magnified
- (C) Virtual, erect, and diminished
- (D) Real, erect, and diminished

33. In Young's Double Slit Experiment, the fringe width β is given by:

- (A) $\lambda d / D$
- (B) $\lambda D / d$
- (C) $D / (\lambda d)$
- (D) $d / (\lambda D)$

34. According to Brewster's law, the refractive index μ of a medium is related to the polarizing angle i_p by:

- (A) $\mu = \sin(i_p)$
- (B) $\mu = \cos(i_p)$
- (C) $\mu = \tan(i_p)$
- (D) $\mu = \cot(i_p)$

35. The energy E of a photon of momentum p is given by (c = speed of light):

- (A) $E = pc$
- (B) $E = p/c$
- (C) $E = pc^2$
- (D) $E = p^2/(2m)$

36. The de Broglie wavelength of an electron accelerated through V volts is approximately:

- (A) $12.27 / \sqrt{V} \text{ \AA}$
- (B) $1.227 / \sqrt{V} \text{ \AA}$
- (C) $12.27 \sqrt{V} \text{ \AA}$
- (D) $12.27 / V \text{ \AA}$

37. In the Bohr model of the hydrogen atom, the radius of the n^{th} orbit is proportional to:

- (A) n
- (B) n^2
- (C) $1/n$
- (D) $1/n^2$

38. The half-life $T_{1/2}$ of a radioactive substance is related to its decay constant λ by:

- (A) $T_{1/2} = \lambda \ln(2)$
- (B) $T_{1/2} = \ln(2) / \lambda$
- (C) $T_{1/2} = \lambda / \ln(2)$
- (D) $T_{1/2} = 1 / \lambda$

39. The maximum binding energy per nucleon occurs for the nucleus of:

- (A) Helium (^4He)
- (B) Oxygen (^{16}O)
- (C) Iron (^{56}Fe)
- (D) Uranium (^{238}U)

40. In a reverse-biased p-n junction, the width of the depletion layer:

- (A) Decreases
- (B) Increases
- (C) Remains the same
- (D) First increases then decreases

41. The logic gate that produces a high (1) output only when all its inputs are high (1) is:

- (A) OR gate
- (B) AND gate
- (C) NOR gate
- (D) NAND gate

42. A Zener diode is primarily used as:

- (A) An amplifier
- (B) A voltage regulator
- (C) A rectifier
- (D) An oscillator

43. If the percentage error in measuring the radius of a sphere is 2%, the percentage error in its volume is:

- (A) 2%
- (B) 4%
- (C) 6%
- (D) 8%

44. The magnitude of the cross product of two parallel vectors is:

- (A) Product of their magnitudes
- (B) Zero
- (C) One
- (D) Infinity

45. The angle of repose on a rough inclined plane is equal to the:

- (A) Angle of incidence
- (B) Angle of friction
- (C) Critical angle
- (D) Angle of deviation

46. When a gas undergoes an adiabatic expansion, its temperature:

- (A) Increases
- (B) Decreases
- (C) Remains constant
- (D) Depends on the specific heat

47. The SI unit of magnetic dipole moment is:

- (A) A m
- (B) A m²
- (C) T m A⁻¹
- (D) Wb m²

48. A particle performs SHM with amplitude A. Its kinetic energy equals its potential energy when its displacement from the mean position is:

- (A) A/2
- (B) A/√2
- (C) A
- (D) Zero

49. The power factor of a purely resistive AC circuit is:

- (A) Zero
- (B) 0.5
- (C) 1
- (D) Infinity

50. The core of an electromagnet is made of soft iron because soft iron has:

- (A) High retentivity and high coercivity
 - (B) Low retentivity and low coercivity
 - (C) High retentivity and low coercivity
 - (D) Low retentivity and high coercivity
-

CATEGORY - II (2 Marks Each)

Only one option is correct. Correct response yields 2 marks. Incorrect response yields -0.5 marks.

51. A solid sphere rolls down two different inclined planes of the same height but different angles of inclination. In both cases, the sphere:

- (A) reaches the bottom with the same speed and in the same time.
- (B) reaches the bottom with the same speed but different times.
- (C) reaches the bottom with different speeds but the same time.
- (D) reaches the bottom with different speeds and different times.

52. A simple pendulum has a time period T_1 on the earth's surface and T_2 at a height R above the earth's surface, where R is earth's radius. The ratio T_2/T_1 is:

- (A) 1
- (B) $\sqrt{2}$
- (C) 2
- (D) 4

53. A parallel plate capacitor is charged by a battery which is then disconnected. A dielectric slab is now inserted between the plates. Which quantity remains constant?

- (A) Capacitance
- (B) Potential difference
- (C) Electric field
- (D) Charge

54. A particle of mass m and charge q is projected with velocity v perpendicular to a uniform magnetic field B . The work done by the magnetic force after one revolution is:

- (A) mv^2/r
- (B) $qvB(2\pi r)$
- (C) Zero
- (D) $(1/2)mv^2$

55. In a Young's double-slit experiment, the intensity of light where the path difference is λ is K . The intensity where the path difference is $\lambda/4$ will be:

- (A) K
- (B) $K/2$
- (C) $K/4$
- (D) Zero

56. A block of mass m is pulled along a horizontal frictionless surface by a thick rope of mass $m/2$. If a force F is applied at the free end of the rope, the force exerted by the rope on the block is:

- (A) F
- (B) $2F/3$
- (C) $F/3$
- (D) $3F/2$

57. Two spheres of the same material have radii R and $2R$. If they are heated to the same temperature and left in identical surroundings, the ratio of their initial rates of cooling is:

- (A) 1:1
- (B) 2:1
- (C) 1:2
- (D) 4:1

58. An ideal gas heat engine operates in a Carnot cycle between 227°C and 127°C . It absorbs 600 kcal at the higher temperature. The amount of heat converted into work is:

- (A) 120 kcal
- (B) 240 kcal
- (C) 480 kcal
- (D) 600 kcal

59. The equivalent resistance between diagonally opposite corners of a wire cube of resistance R on each edge is:

- (A) $5R/6$
- (B) $7R/12$
- (C) $3R/4$
- (D) R

60. The magnetic susceptibility of a paramagnetic substance at -73°C is 0.0060. Its susceptibility at -173°C will be:

- (A) 0.0030
- (B) 0.0120
- (C) 0.0180
- (D) 0.0045

61. In a series LCR circuit connected to an AC source, the voltage across R , L , and C is 10 V each. If the capacitor is short-circuited, the new voltage across the inductor will be:

- (A) 10 V
- (B) $10/\sqrt{2}$ V
- (C) $10/3$ V
- (D) 0

62. If the momentum of an electron is changed by p , the de Broglie wavelength associated with it changes by 0.5%. The initial momentum of the electron is:

- (A) $100 p$
- (B) $200 p$
- (C) $400 p$
- (D) $p/200$

63. A radioactive sample with half-life T contains N_0 active nuclei at $t=0$. The number of nuclei decayed in time $t = T/2$ is:

- (A) $N_0 / \sqrt{2}$
- (B) $N_0 (1 - 1/\sqrt{2})$
- (C) $N_0 / 2$
- (D) $N_0 / 4$

64. Two thin lenses of focal lengths f_1 and f_2 are in contact and coaxial. The power of the combination is:

- (A) $f_1 + f_2$
- (B) $(f_1 f_2) / (f_1 + f_2)$
- (C) $1/f_1 + 1/f_2$
- (D) f_1 / f_2

65. The minimum deviation produced by a hollow prism filled with a liquid is 30° . If the refracting angle of the prism is 60° , the refractive index of the liquid is:

- (A) 1.5
- (B) 1.33
- (C) 1.414
- (D) 1.61

CATEGORY - III (2 Marks Each)

One or more options may be correct. Correct response yields 2 marks. No negative marking.

66. A satellite is orbiting the earth in a circular orbit. If its mass is suddenly doubled without changing its orbital speed or radius, which statements are correct?

- (A) Its time period will remain unchanged.
- (B) Its kinetic energy will be doubled.
- (C) Its potential energy will remain unchanged.
- (D) Its angular momentum will be doubled.

67. Which of the following equations represent a traveling wave?

- (A) $y = A \sin(kx - \omega t)$
- (B) $y = A \cos(kx + \omega t)$
- (C) $y = A \sin(kx) \cos(\omega t)$
- (D) $y = A e^{-(x - vt)^2}$

68. For a photoelectric effect experiment, which of the following statements are true?

- (A) The stopping potential depends on the intensity of incident light.
- (B) The stopping potential depends on the frequency of incident light.
- (C) The saturation current depends on the intensity of incident light.
- (D) The photoelectric emission is an instantaneous process.

69. In an LCR series circuit connected to an AC source at resonance:

- (A) The impedance of the circuit is minimum.
- (B) The current in the circuit is maximum.
- (C) The voltage across the inductor and capacitor are equal and opposite in phase.
- (D) The power factor of the circuit is zero.

70. Let V and E denote the gravitational potential and field at a point, respectively. It is possible that:

- (A) $V = 0$ and $E = 0$
- (B) $V = 0$ and $E \neq 0$
- (C) $V \neq 0$ and $E = 0$
- (D) $V \neq 0$ and $E \neq 0$

71. A particle moves in a circular path with continuously increasing speed. Which of the following are correct?

- (A) Its acceleration vector is strictly perpendicular to its velocity vector.
- (B) Its acceleration vector is not perpendicular to its velocity vector.
- (C) The work done by the net force is positive.
- (D) Its angular momentum about the center is conserved.

72. In an isolated charged capacitor with a dielectric:

- (A) If the dielectric is removed, the potential difference increases.
- (B) If the dielectric is removed, the energy stored increases.
- (C) If the dielectric is removed, the charge increases.
- (D) If the dielectric is removed, the electric field decreases.

73. Which of the following quantities are quantized in Bohr's model of a hydrogen atom?

- (A) Linear momentum
- (B) Angular momentum
- (C) Kinetic energy
- (D) Total energy

74. A ray of light is incident from a denser medium to a rarer medium. The critical angle is θ_c .

- (A) If angle of incidence $i > \theta_c$, total internal reflection occurs.
- (B) If $i < \theta_c$, the ray is partially reflected and partially refracted.
- (C) The refractive index of denser medium with respect to rarer medium is $1/\sin(\theta_c)$.
- (D) Total internal reflection can occur when light travels from rarer to denser medium.

75. In the decay process: $A \rightarrow B + \alpha$, $B \rightarrow C + \beta$:

- (A) Mass number of A is 4 more than that of C.
- (B) Atomic number of A is 1 more than that of C.
- (C) A and C are isotopes.
- (D) B and C are isobars.

ANSWER KEY

Category - I

1. (B) 2. (A) 3. (A) 4. (A) 5. (A) 6. (C) 7. (A) 8. (A) 9. (B) 10. (A)
11. (C) 12. (C) 13. (B) 14. (C) 15. (B) 16. (A) 17. (A) 18. (B) 19. (B) 20. (B)
21. (A) 22. (C) 23. (B) 24. (B) 25. (B) 26. (C) 27. (D) 28. (B) 29. (A) 30. (A)
31. (D) 32. (C) 33. (B) 34. (C) 35. (A) 36. (A) 37. (B) 38. (B) 39. (C) 40. (B)
41. (B) 42. (B) 43. (C) 44. (B) 45. (B) 46. (B) 47. (B) 48. (B) 49. (C) 50. (B)

Category - II

51. (B) 52. (C) 53. (D) 54. (C) 55. (B) 56. (B) 57. (B) 58. (A) 59. (A) 60. (B)
61. (B) 62. (B) 63. (B) 64. (C) 65. (C)

Category - III

- 66. (A, B, D)
- 67. (A, B, D)
- 68. (B, C, D)
- 69. (A, B, C)
- 70. (A, B, C, D)
- 71. (B, C)
- 72. (A, B)
- 73. (B, C, D)
- 74. (A, B, C)
- 75. (A, B, D)